String s_1 = "shraddha"

String s_2 = "koli"

String name = s_1 + s_2

shraddha + koli

Shraddha koli

Made with Goodnotes

9)

```
public class Main{
public static void main(String[] args) {
      String str1="NPTEL";
      String str2="java";
      int a=20;
      int b=10;
      System.out.println(str1+a+b); //Statement 1

      System.out.println(a+b+str2); //Statement 2
    }
}
```

Which of the following statement(s) is/are correct?

a. The output of the Statement 1 is NPTEL2010
b. The output of the Statement 1 is NPTEL30
c. The output of the Statement 2 is 1020java
d. The output of the Statement 2 is 30java

NPTL2010

x + y

↓

String → Concatⁿ

x + y + string

↓, Addⁿ

str1 + a + b

NPTL2010

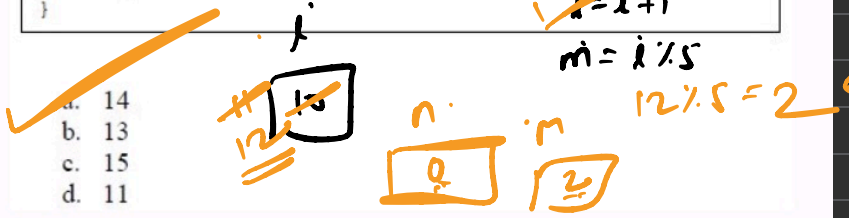
Made with Goodnotes

1) **What is the output of the following program?**

```
public class Question
{
      public static void main(String[] args) {

         int i = 10;
         int n = i++%5;
         int m = ++i%5;
          System.out.println(i+n+m);
       }
}
```

a. 14
b. 13
c. 15
d. 11



Made with Goodnotes

6) **Consider the following program.**

```
public class Question{
      public static void main(String args[]){
            for(int a=1;a<2;a+=1){
                  System.out.print(a+++a);
            }
      }
}
```

a + +a

a++ +a

What will be the output of the program if it is executed?

a. 3
b. 2
c. 4
d. 1

Made with Goodnotes

CodeGreedy



Arrays in JAVA



CodeGreedy

“Success doesn’t come from what you do occasionally, it comes from what you do consistently.”

Made with Goodnotes

WHY DO WE NEED ARRAYS

CodeGreedy

Imagine you want to store marks of 5 students:

```
int m1 = 90;
int m2 = 85;
int m3 = 78;
int m4 = 92;
int m5 = 88;
```

Managing 100 different variables becomes difficult.



x = 100, 1000

What if there are 1000 students?

Made with Goodnotes

int $x_1 = 10$

int $x_2 = 20$

int $x_3 = 30$

$arr[] = \{ 10, 20, 30 \}$

$arr[] = \{ x_1, x_2, x_3 \}$

Made with Goodnotes

WHAT IS ARRAY

CodeGreedy

Collection of same data type stored in contiguous memory

syntax

```
dataType[] arrayName = new dataType[size]
```

Example

```
int[] arr = new int[5];
```



CodeGreedy

Made with Goodnotes

int arr[5]

} CPP / C

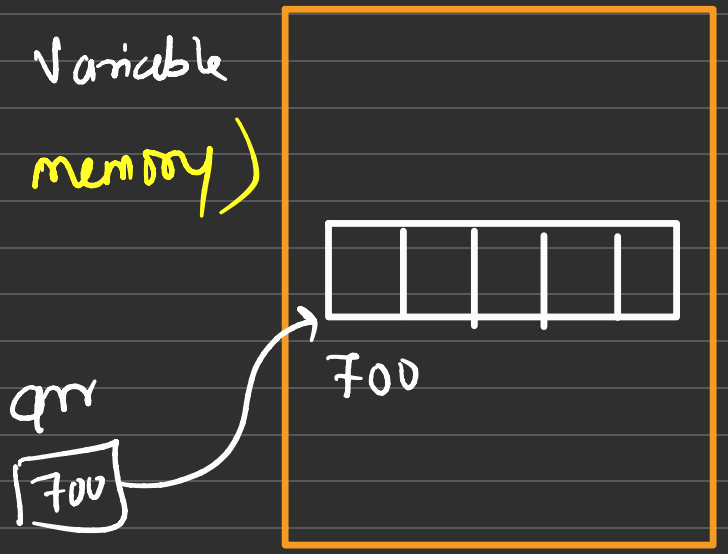


---

int []arr ⟶ Refrence Variable

(points to memory)

arr = new int[5]



Made with Goodnotes

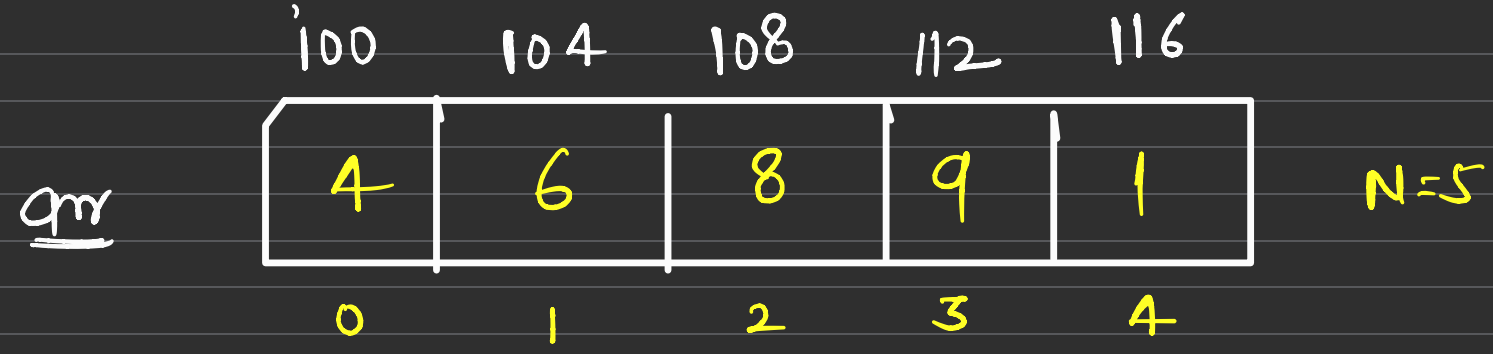
int [] arr; —— (Reference variable)

arr = new int[size]

1. int [] arr = new int[10]

2. int [] brr = { 10, 20, 30, 40 }

Made with Goodnotes



arr(2)

arr + 2

100 + 2x4

100 + 8

108

Made with Goodnotes

4B

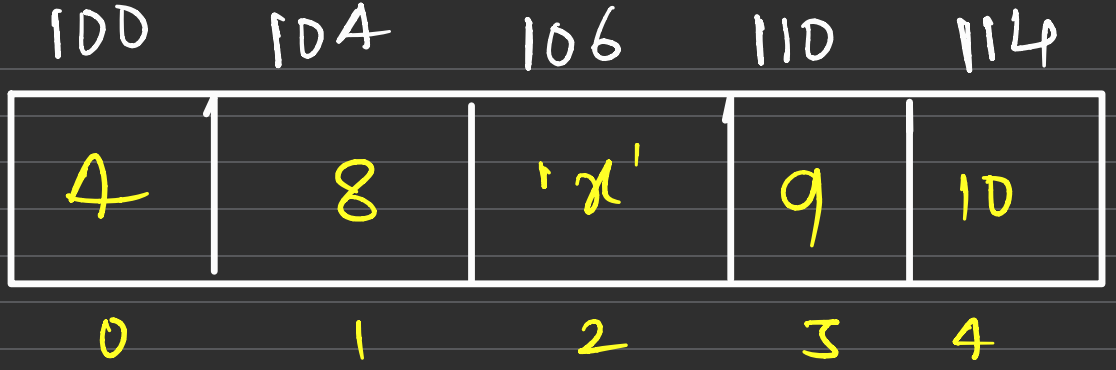


1B (C/CPP) ASCII (-128 to 127)
(0 to 255)

2B (Java) UNICODE ← 0 to 255 (ASCII)

16 bits 0 to 65535

Made with Goodnotes



Made with Goodnotes

TYPES OF ARRAY

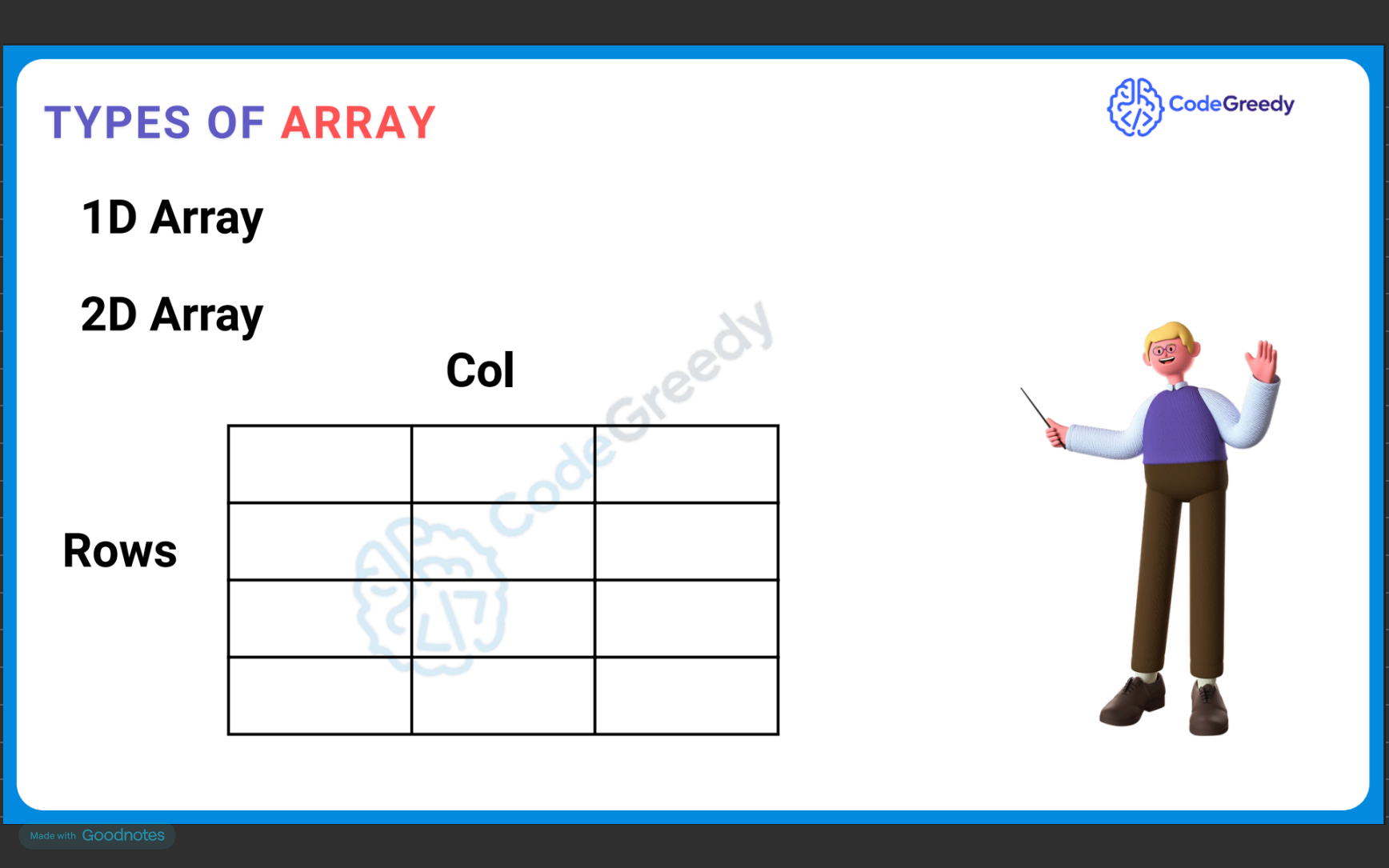
CodeGreedy

1D Array

2D Array

Col

Rows



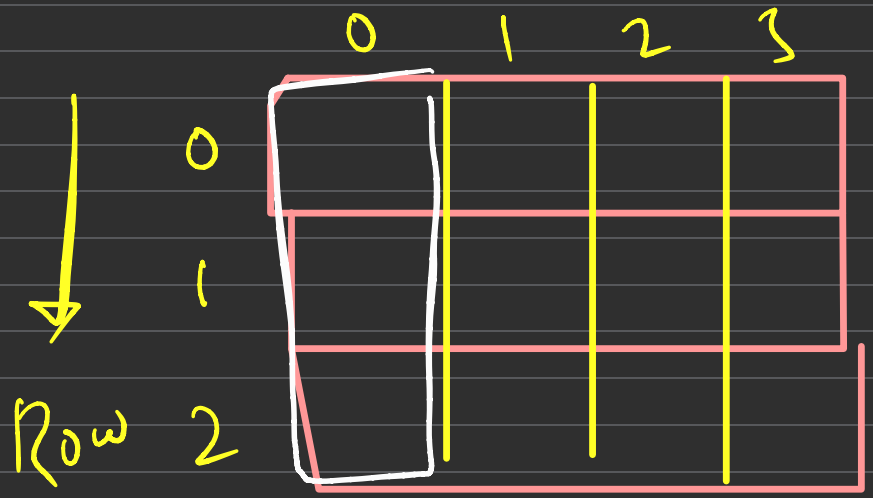
Made with Goodnotes

```
int [] arr = new int [6]
int [][] mat = new int [3][4]
```

Row

↳ Column

⟶ Col

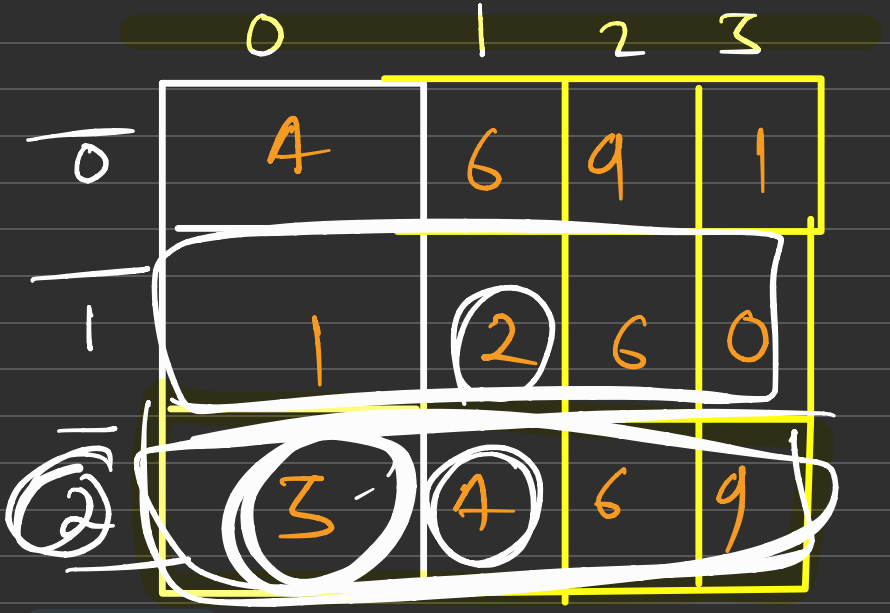


arr(3)

1

Made with Goodnotes

3	1	0	6	2
0	1	2	3	4




```
for (i=0 ; i<3 ; i++)
{
    for (j=0 ; j<4 ; j++)
    {
        print (a[i][j])
    }
}
```

Made with Goodnotes

INTERVIEW QUESTION

CodeGreedy

Why array always start from index 0



Made with Goodnotes

TRAVERSING ARRAYS

CodeGreedy

Using for loop

```
for(int i = 0; i < arr.length; i++) {
   System.out.println(arr[i]);
}
```

for-each loop

```
for(int num : arr) {
   System.out.println(num);
}
```



CodeGreedy

Made with Goodnotes

BASIC OPERATIONS ON ARRAYS

CodeGreedy

insert

```
arr[2] = 100;
```

Update

```
arr[1] = 50;
```

Earlier value get replaced by 50.



CodeGreedy

Made with Goodnotes

COMMON INTERVIEW PROBLEMS

CodeGreedy

Find Maximum number from the array

Reverse the array

Check whether given number exist in the array.



Made with Goodnotes

1. Two Sum

Solved

Easy | Topics | Companies | Hint | Re-do

You are given an array of integers `nums` and an integer `target`, return *indices of the two numbers such that they add up to* `target`.

You may assume that each input would have ***exactly* one solution**, and you may not use the *same* element twice.

You can return the answer in any order.

Example 1:

Input: `nums = [2,7,11,15], target = 9`

Output: `[0,1]`

Explanation:

Made with Goodnotes

283. Move Zeroes

Solved

Easy | Topics | Companies | Hint | Re-do

Given an integer array `nums`, move all `0`'s to the end of it while maintaining the relative order of the non-zero elements.

Note that you must do this in-place without making a copy of the array.

Example 1:

```
Input: nums = [0,1,0,3,12]
Output: [1,3,12,0,0]
```

Example 2:

```
Input: nums = [0]
Output: [0]
```

Made with Goodnotes

121. Best Time to Buy and Sell Stock

Solved

Easy | Topics | Companies | Re-do

You are given an array `prices` where `prices[i]` is the price of a given stock on the ith day.

You want to maximize your profit by choosing a **single day** to buy one stock and choosing a **different day in the future** to sell that stock.

Return *the maximum profit you can achieve from this transaction*. If you cannot achieve any profit, return `0`.

Example 1:

```
Input: prices = [7,1,5,3,6,4]
Output: 5
Explanation: Buy on day 2 (price = 1) and sell on day 5
(price = 6), profit = 6-1 = 5.
Note that buying on day 2 and selling on day 1 is not
allowed because you must buy before you sell.
```

Made with Goodnotes

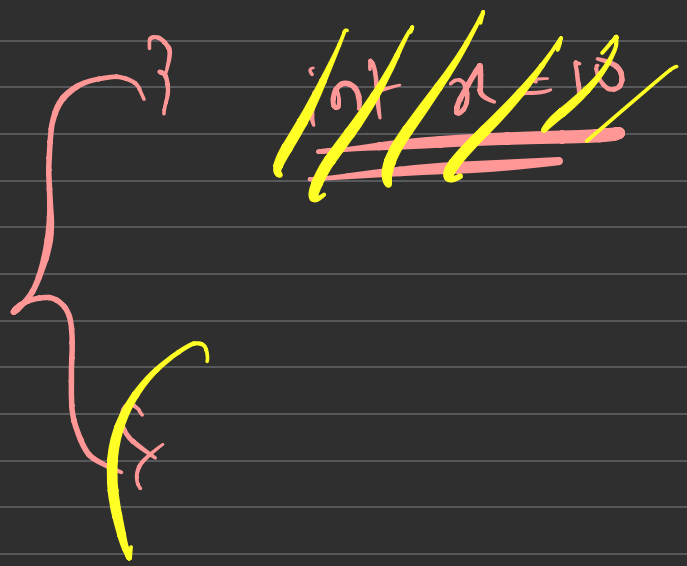
Storage classes

```
printf(Hello)
x = x++
printf(x)
```

Hello

Syntax for Var

Made with Goodnotes



Made with Goodnotes